Bean consumption is associated with greater nutrient intake, reduced systolic blood pressure, lower body weight, and a smaller waist circumference ...

BACKGROUND: Epidemiological studies have shown positive findings associated with legume consumption and measures of cardiovascular disease and obesity. However, few observational trials have examined beans as a separate food variable when determining associations with health parameters. OBJECTIVE: To determine the association of consuming beans on nutrient intakes and physiological parameters using the National Health and Examination Survey (NHANES) 1999-2002. METHODS: Using data from NHANES 1999-2002, a secondary analysis was completed with a reliable 24-hour dietary recall where three groups of bean consumers were identified (N = 1,475). We determined mean nutrient intakes and physiological values between bean consumers and non-consumers. Least square means, standard errors and ANOVA were calculated using appropriate sample weights following adjustment for age, gender, ethnicity and energy. RESULTS: Relative to non-consumers, bean consumers had higher intakes of dietary fiber, potassium, magnesium, iron, and copper ($p's < 0.05$). Those consuming beans had a lower body weight ($p = 0.008$) and a smaller waist size ($p = 0.043$) relative to non-consumers. Additionally, consumers of beans had a 23% reduced risk of increased waist size ($p = 0.018$) and a 22% reduced risk of being obese ($p = 0.026$). Also, baked bean consumption was associated with a lower systolic blood pressure. CONCLUSIONS: Bean consumers had better overall nutrient intake levels, better body weights and waist circumferences, and lower systolic blood pressure in comparison to non-consumers. These data support the benefits of bean consumption on improving nutrient intake and health parameters.